

Manoj Sachdev

professor

基本信息

姓名: Manoj Sachdev

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邮箱: msachdev@uwaterloo.ca

地区: Canada

年龄范围: 50岁以上

海外经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可, 展现了卓越的研究能力和影响力。

学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

低功耗电路与能量管理

存储器设计与架构

数字集成电路设计

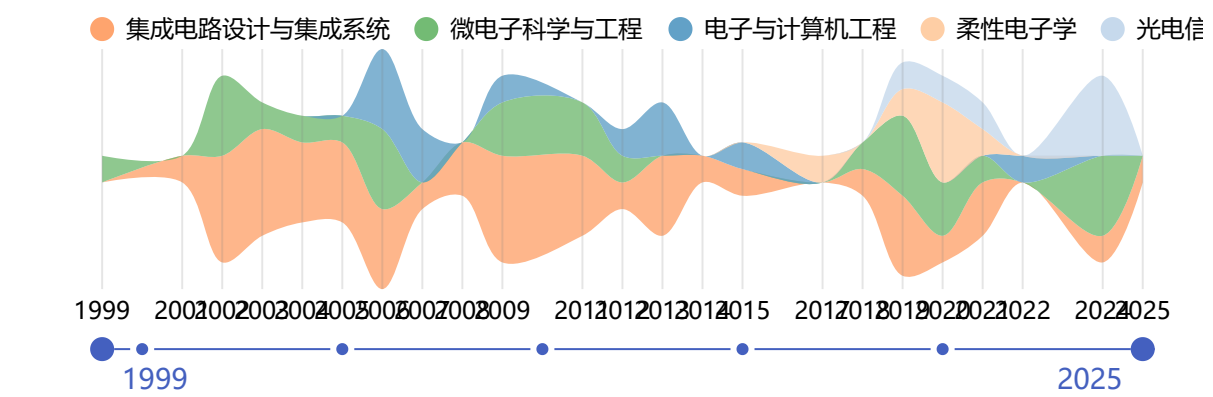
微电子器件与制造

个人简介

Dr. Manoj Sachdev is a Professor in the Department of Electrical and Computer Engineering at the University of Waterloo. His research focuses on digital and mixed - signal circuit design, and IC test and manufacturing. He has over 180 publications, five books, and 30+ US patents. He is an IEEE and Engineering Institute of Canada Fellow, and serves on editorial and program committees. He holds a doctorate from Brunel University and a bachelor's from the University of Roorkee. He has won awards, taught courses like ECE 445 and ECE

637, and has many recent publications. He is currently considering graduate student applications. Contact: msachdev@uwaterloo.ca, E5 4027, 519 - 888 - 4567 x33370.

近年发表的论文



Design and Optimization of a Gold and Silver Nanoparticle-Based SERS Biosensing Platform

SENSORS

Saha, Soumyadeep; Sachdev, Manoj; Mitra, Sushanta K.

被引用次数: 0 2025

VLFF - A very low-power flip-flop with only two clock transistors

INTEGRATION-THE VLSI JOURNAL

Maheshwari, Yugal Kishore; Sachdev, Manoj

被引用次数: 0 2025

Leveraging Banking and Charge-Sharing schemes for Low-Power Micro-displays

2024 IEEE CANADIAN CONFERENCE ON ELECTRICAL AND COMPUTER ENGINEERING, CCECE 2024

Ranjan, Shubham; Gohardehi, Sheida; Sachdev, Manoj

被引用次数: 0 2024

A power-efficient, single-phase, contention-free flip-flop with only three clock transistors

MICROELECTRONICS JOURNAL

Maheshwari, Yugal Kishore; Sachdev, Manoj

被引用次数: 1 2024

Energy Restoration in Data Drivers for Low-Power Digitally Driven OLEDoS Microdisplays

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Gohardehi, Sheida; Sachdev, Manoj

被引用次数: 0 2024

Low-Power Row Driver for Digitally Driven OLEDoS Microdisplay

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

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被引用次数: 1

2024

Helimagnet-based nonvolatile multi-bit memory units

APPLIED PHYSICS LETTERS

Islam, Rabiul; Li, Peng; Beg, Marijan; Sachdev, Manoj; Miao, Guo-Xing

被引用次数: 2

2023

Recent advances in label-free optical, electrochemical, and electronic biosensors for glioma biomarkers

BIOMICROFLUIDICS

Saha, Soumyadeep; Sachdev, Manoj; Mitra, Sushanta K.

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2023

Enhancing Strong PUF Security With Nonmonotonic Response Quantization

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Stangherlin, Kleber; Wu, Zhuanhao; Patel, Hiren; Sachdev, Manoj

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2022

Enhanced Surface-Emitting Vertical InGaN Light-Emitting Diodes by Self-Aligned Sidewall Electrodes and Optical Cavity Length Modulation

ACS APPLIED ELECTRONIC MATERIALS

Asad, Mohsen; Li, Qing; Sachdev, Manoj; Wong, William S.

被引用次数: 2

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Energy-Efficient Full-Swing Logic Circuits With Unipolar TFTs on Flexible Substrates

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Qing; Lee, Czung-Ho; Asad, Mohsen; Wong, William S.; Sachdev, Manoj

被引用次数: 5

2021

A 350 mV, 2 MHz, 16-kb SRAM with programmable wordline boosting in the 65 nm CMOS technology

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Nabavi, Morteza; Sachdev, Manoj

被引用次数: 2

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Body Biased Sense Amplifier With Auto-Offset Mitigation for Low-Voltage SRAMs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Patel, Dhruv; Neale, Adam; Wright, Derek; Sachdev, Manoj

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CMOS-Like Logic Circuits With Unipolar Thin-Film Transistors on Flexible Substrate

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2020

A low-power low-offset charge-sharing technique for double-tail comparators

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Khorami, Ata; Saeidi, Roghayeh; Sachdev, Manoj

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2020

Monitoring Aging Defects in STT-MRAMs

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Radhakrishnan, Govind; Yoon, Youngki; Sachdev, Manoj

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Thermal and optical properties of high-density GaN micro-LED arrays on flexible substrates (vol 73, 104724, 2020)

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Thermal and optical properties of high-density GaN micro-LED arrays on flexible substrates

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SRAM Dosimeter for Characterizing the TRIUMF Proton and Neutron Beams

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A Charge-Depletion Study of an Electrostatic Generator With Adjustable Output Voltage

IEEE SENSORS JOURNAL

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Size-dependent optoelectrical properties of 365 nm ultraviolet light-emitting diodes

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Hybrid Latch-Type Offset Tolerant Sense Amplifier for Low-Voltage SRAMs

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Patel, Dhruv; Neale, Adam; Wright, Derek; Sachdev, Manoj

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A 290-mV, 3.34-MHz, 6T SRAM With pMOS Access Transistors and Boosted Wordline in 65-nm CMOS Technology

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Nabavi, Morteza; Sachdev, Manoj

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2018

An nMOS Static ESD Power Supply Clamp With Thyristor Delay Element and 180 pA Leakage in 65 nm CMOS Technology

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Effect of Mechanical Strain on Hydrogenated Amorphous Silicon Thin-Film Transistors and Compensation Circuits on Flexible Substrates

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Neutron Radiation Induced Soft Error Rates for an Adjacent-ECC Protected SRAM in 28 nm CMOS

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2016

Adjacent-MBU-Tolerant SEC-DED-TAEC-yAED Codes for Embedded SRAMs

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A 32 kb Macro with 8T Soft Error Robust, SRAM Cell in 65-nm CMOS

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A 167-ps 2.34-mW Single-Cycle 64-Bit Binary Tree Comparator With Constant-Delay Logic in 65-nm CMOS

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Sharifkhani, Mohammad; Rahiminejad, Ehsan; Jahinuzzaman, Shah M.; Sachdev, Manoj

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SRAM Cell Stability: A Dynamic Perspective

IEEE JOURNAL OF SOLID-STATE CIRCUITS

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Jahinuzzaman, Shah M.; Shah, Jaspal Singh; Rennie, David J.; Sachdev, Manoj

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A Low-Power Ternary CAM With Positive-Feedback Match-Line Sense Amplifiers

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Mohan, Nitin; Fung, Wilson; Wright, Derek; Sachdev, Manoj

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A Soft Error Tolerant 10T SRAM Bit-Cell With Differential Read Capability

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Low-Leakage Storage Cells for Ternary Content Addressable Memories

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

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2009

An Analytical Model for Soft Error Critical Charge of Nanometric SRAMs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Jahinuzzaman, Shah M.; Sharifkhani, Mohammad; Sachdev, Manoj

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A new flip-flop-based transient power supply clamp for ESD protection

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Sarbishaei, Hossein; Semenov, Oleg; Sachdev, Manoj

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A 5-Gb/s CDR circuit with automatically calibrated linear phase detector

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Kok-Siang, Tan; Sulaiman, Mohd-Shahiman; Reaz, Mamun; Hean-Teik, Chuah; Sachdev, Manoj

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Low-capacitance and charge-shared match lines for low-energy high-performance TCAMs

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Design of high-speed clock and data recovery circuits

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Segmented virtual ground architecture for low-power embedded SRAM

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Novel gate and substrate triggering techniques for deep sub-micron ESD protection devices

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Thermal runaway in integrated circuits	
IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY	
Vassighi, Arman; Sachdev, Manoj	
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Impact of self-heating effect on long-term reliability and performance degradation in CMOS circuits	
IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY	
Semenov, O; Vassighi, A; Sachdev, M	
被引用次数: 137	2006
DTMOS technique for low-voltage analog circuits	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Maymandi-Nejad, Mohammad; Sachdev, Manoj	
被引用次数: 21	2006
Design techniques and test methodology for low-power TCAMs	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Mohan, Nitin; Fung, Wilson; Wright, Derek; Sachdev, Manoj	
被引用次数: 43	2006
Weak cell detection in deep-submicron SRAMs: A programmable detection technique	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Pavlov, Andrei; Sachdev, Manoj; Pineda de Gyvez, Jose	
被引用次数: 9	2006
Design of a 1.7-GHz low-power delay-fault-testable 32-b ALU in 180-nm CMOS technology	
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Chatterjee, B; Sachdev, M	
被引用次数: 5	2005
A monotonic digitally controlled delay element	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
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Designing leakage tolerant, low power wide-OR dominos for sub-130 nm CMOS technologies	
MICROELECTRONICS JOURNAL	
Chatterjee, B; Sachdev, M; Krishnamurthy, R	
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Docking, S; Sachdev, M	
被引用次数: 41	2004
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IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY	
Vassighi, A; Semenov, O; Sachdev, M; Keshavarzi, A; Hawkins, C	
被引用次数: 28	2004
A low-power reduced swing global clocking methodology	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Asgari, FHA; Sachdev, M	
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A DFT technique for testing high-speed circuits with arbitrarily slow testers	
JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS	
Nummer, M; Sachdev, M	
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A digitally programmable delay element: Design and analysis	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Maymandi-Nejad, M; Sachdev, M	
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Effect of CMOS technology scaling on thermal management during burn-in	
IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING	
Semenov, O; Vassighi, A; Sachdev, M; Keshavarzi, A; Hawkins, CF	
被引用次数: 21	2003
Testing high-performance pipelined circuits with slow-speed testers	
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Nummer, M; Sachdev, M	
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Impact of gate induced drain leakage on overall leakage of submicrometer CMOS VLSI circuits	
IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING	
Semenov, O; Pradzynski, A; Sachdev, M	
被引用次数: 33	2002
Continuous time common mode feedback technique for sub 1 V analogue circuits	
ELECTRONICS LETTERS	
Maymandi-Nejad, M; Sachdev, M	
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Impact of technology scaling on thermal behavior of leakage current in sub-quarter micron MOSFETs: perspective of low temperature current testing	
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A comparative analysis of low-power low-voltage dual-edge-triggered flip-flops

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Chung, W; Lo, T; Sachdev, M

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Design for delay testability in high-speed digital ICs

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Kerkhoff, HG; Speek, H; Shashani, M; Sachdev, M

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2001

Off-chip diagnosis of aperture jitter in full-flash analog-to-digital converters

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Rosing, R; Kerkhoff, H; Tangelder, R; Sachdev, M

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1999

Monitoring Aging Defects in STT-MRAMs

TCAD

Govind Radhakrishnan; Youngki Yoon; Manoj Sachdev

被引用次数: 0

学术合作



Adam Neale	合作次数: 5
滑铁卢大学	
Derek Wright	合作次数: 2
滑铁卢大学	
Dhruv Patel	合作次数: 2
多伦多大学	
David Nairn	合作次数: 2
滑铁卢大学	
Zhuanhao Wu	合作次数: 1
滑铁卢大学	

还有 4 个合作者